

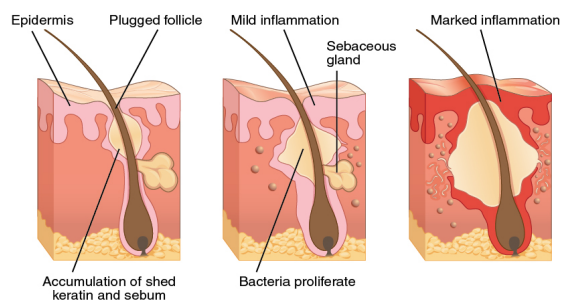
Beyond the surface: Breaking Down the Physical and Psychological Impacts of Acne Vulgaris

How Genetics, Hormones and Daily Habits influence Our Skin and What it Means for Our Mental Wellbeing

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Cross-section diagram showing healthy hair follicle vs. clogged pores, sebaceous gland, sebum buildup, and bacterial acne lesion progression (Lumen Learning: Disorders of Skin)

Abstract

Acne vulgaris is a skin condition that affects 85% of adolescents aged between 12-24 years old. Acne is not only a physical skin condition, but it can also build into psychological distress. This overview dives deep into the root causes of common acne, from sebum production to lifestyle and habits, while also breaking down symptoms, severities, and different treatment pathways. This overview connects the biology of acne with the emotional health aspect to help readers seek early treatment and feel good in their skin again.

Introduction

Imagine standing in a classroom, a coffee shop with a friend, or even a meeting, trying to learn or laugh. Instead, you're following people's eye contact and body language, trying to see if they notice your acne or scarring. If this situation seems familiar, you are not alone. Acne vulgaris affects around 9% of the world's population, 85% of which are between 12-24 years old, which means this is one of the most common skin conditions in the world (Tan J.K.L., Bhate K 2015).

Acne vulgaris (also known as common acne) is a chronic inflammatory disease of the hair follicle-sebaceous gland (the skin's natural pairing of a hair follicle and its attached oil gland). This skin condition appears when the hair follicles are blocked with excess sebum (oil) and dead skin cells, creating inflammation and breakouts.

This skin disease affects people not only physically, but also carries an emotional weight. Since acne vulgaris is a visible skin disease, one's self esteem and confidence can easily be lost. For most individuals who do not struggle with acne, actions like leaving the house seem simple, but for those who do struggle, leaving the house can seem overwhelming by thinking about who they're going to see or how much of their acne they will have to cover up.

The root causes of common acne that we will be discovering today are hormonal changes, excess sebum production (excess oil), inflammation, genetics and lifestyle and environmental triggers. All of these causes can induce or aggravate this skin condition. The development of acne involves:

Root causes

Hormonal triggers & Excess sebum production: When locally produced or circulating androgen hormones (ex: testosterone) attach to the androgen receptors in the sebaceous cell (oil gland cell), oil production is triggered. The excess oil gets trapped in the pores which causes inflammation. This process is especially active during hormone imbalance, pregnancy, fluctuates during the menstrual cycle and puberty (Zoubilis C.C., Degitz K. 2004).

Genetics: Genetics contributes to approximately 81% of the risk towards progress in this skin disease (Bataille V et al 2002). Although there is not a specific acne gene, it is considered a polygenic (disease that is influenced by multiple genes) condition that is

inherited. The trait inherited dictates the oil production, skin cell shedding and inflammatory responses (Zhang H, Zhang Z 2023).

Lifestyle & Environmental triggers: One's environment and daily activities heavily affect acne vulgaris. Lifestyle habits that can trigger acne vulgaris are high sugar foods such as sugars, processed carbohydrates, white bread, and dairy products that trigger your insulin levels to spike resulting in excess oil production and inflammation. Poor hygiene (ex: not washing off sweat and germs such as cutibacterium acnes (C. acnes)), allowing sweat, dirt, and oils to rest on your skin after a workout or high intensity activity will quickly result in clogged pores. Stress and lack of sleep, elevate your cortisol (stress hormone). Elevated cortisol causes excessive sebum production and aggravates acne. Environmental factors include humidity, overexposure to the sun and to pollution also plays a significant role in triggering breakouts (Deng Y, Wang F, He L 2024).

Signs & Symptoms of Acne Vulgaris:

The two main categories of acne are inflammatory and non-inflammatory. Inflamed acne presents itself with papules (small and solid raised bump), pustules (raised bump that contains visible pus), and cysts (sacs filled with fluid or semi solid substances)(Eichenfield D.Z., Sprague J, Eichenfield L.F. 2021 ; Fox L et al 2016). Inflammatory acne occurs when there is too much excess sebum in the blocked hair follicles causing inflammation. The symptoms of non-inflammatory acne are open comedones (blackheads) and closed comedones (whiteheads). These symptoms occur because the hair follicle is clogged with dead skin cells and excess oil. Identifying the type of acne you might have is important to start the treatment pathway:

Non-inflammatory: To identify non-inflammatory acne vulgaris symptoms we need to observe the open and closed comedones. An open comedone means that the pores are dilated (open) and are being exposed. Open comedones are more commonly referred to as "blackheads". Blackheads are visibly black because there are trapped dead skin cells and oil in your pores. This trapped mixture then reacts to oxygen causing the mixture to "oxidize" causing the trapped material to change colors. This process is similar to when fruits like apples or bananas turn brown after sitting out in open air. "Whiteheads" can also be a symptom of non-inflammatory acne. Whiteheads are closed comedones meaning they stay under a thin layer of your skin. This allows

the oil to be protected from open air, resulting in having a white appearance (Wong C.M. et al 2024).

Inflamed acne: To identify inflammatory acne vulgaris symptoms we need to observe if papules, pustules, nodules and cysts are present (Zaenglein A.L et al 2016). Papules are small, firm, bumps with a “red border”. Often, they are tender when touched. Pustules are commonly known as pimples. They are developed directly on top of papules. They are fluid filled with a white or sometimes yellow hue pus. Nodules are large and painful lumps that go deep past the outer surface of the skin. Nodules do not have a “head” unlike pustules or whiteheads. They are rather firm and very inflamed. Cysts are deep fluid filled bumps that are very painful and are highly common to cause scarring.

Diagnosing Acne Vulgaris

Diagnosing acne vulgaris is commonly done by a clinical examination. The doctor will look for acne characteristics such as open comedones (blackheads), pustules, inflamed papules, closed comedones (whiteheads), and nodules and cysts. Depending on which of the acne vulgaris characteristics are present, acne is classified into three different severity categories: mild, moderate and severe. These classifications serve as a guide to decide which treatment pathways are the most appropriate (Eichenfield L.F. et al 2013).

Note: Symptoms vary widely between individuals; readers should not self-diagnose based on this list. If someone suspects they have this condition, they should consult a healthcare provider.

Treating and Managing Acne Vulgaris:

Depending on the severity of acne vulgaris there are many different treatment options such as topical therapies (tretinoin), benzoyl peroxide, azelaic acid (topical medication that reduces inflammation), or combinations of topical treatments. Mild acne can be managed using these fixed-dose topicals. These treatments help the clogged hair follicles and surface-level characteristics. According to the National Center for Biotechnology Information, mixing topical treatments like benzoyl peroxide with a retinoid such as tretinoin (a Vitamin A medication) can be more effective than either of

the treatments alone. It is important to note that patients should always follow the doctor's recommendation instead of trying treatments spontaneously.

When the severity of acne is between moderate and severe, treatment progresses from topical skincare to oral medications and treatments. These oral medications help to reduce inflammation that the topical skincare cannot treat alone. For women, sometimes hormonal medication is prescribed, such as contraceptive pills or spironolactone (medication that reduces the effects of androgens in some women) which help regulate sebum production mediated by androgens (Reynolds R.V. et al 2024).

When acne is severe and involves scarring, nodules & cysts, or treatment-resistant acne (acne that fails to clear up after standard treatments), then oral isotretinoin is the most effective treatment pathway. Since oral isotretinoin has a high success rate, the guidelines advise to start the medication as early as possible to minimize permanent scarring (Keow S, Xiong G, Abu-Hilal M 2025).

Living with Acne Vulgaris

Severe acne vulgaris can often be painful due to the inflamed nodules and cysts (Hazarika, N. & Archana M, 2016). Since nodules and cysts are deep in our skin layers, they create pressure and inflammation making it very uncomfortable for simple daily facial movements such as speaking, chewing or even smiling. In addition, nodules and cysts may lead to permanent scarring (Zhou C et al 2023). Scarring can cause psychosocial morbidity such as depression and suicidality (Xu S et al 2021).

Physical facial changes such as acne and scarring can trigger depression, social detachment and suicidality (Gupta N et al 2023). Understanding that the struggles of acne are not only physical but also have a huge emotional toll is critical (Hazarika, N. & Archana M, 2016). Early treatments will help prevent permanent scarring and the mental health impacts. (Zhou C et al 2023).

Dealing with acne vulgaris goes far beyond the surface of the skin and deliberately impacts mental wellbeing. The continuous presence of breakouts can feed into personal insecurities and lead to social withdrawal. Recognizing the emotional toll acne vulgaris is vital for treatment approaches, as reducing these wellbeing burdens is just as important as clearing physical symptoms.

Emphasize: Treatment plans are highly individualized; readers should always consult their personal physician before starting or modifying treatment.

Conclusion

Acne vulgaris is one of the most common skin diseases, affecting 9% of the world's population in a physical and mental manner. Luckily, there now have been many studies and new treatments to make acne more manageable. Through ongoing clinical research, there is an advancement in treatments, holistic care and knowledge on acne vulgaris allowing us to have a bigger picture of what this disease is and also a wider range of treatment options. With the resources we have accessible today, early clinical treatments allow effective solutions, helping patients manage their acne, minimize scarring but also to feel more confident, empowered and able to take back control of their quality of life.

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All claims in this article are supported by evidence-based sources.

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